Technical Note

Power Electronics and Control Systems

Parallel grid-forming modules

power sharing, loss of the main grid, earthing and common mode

Design note of the model ***pgfm_ctrl_analysis*** *(**grid-forming**, folder **parallel_of_GFMs**): the model as it runs today, what the literature offers, the first control version, implemented in the library block and tested on a companion model, the common mode of modules on one battery and the earthing of an industrial battery storage system.*

Authors:	**Revision:**	**Date:**
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Abstract: The model `pgfm_ctrl_analysis` puts three 250 kW active front ends in parallel on one lithium-ion battery and on a 690 V grid emulator. Its purpose is the power sharing among grid-forming modules: in normal operation the modules follow the set-points of the plant; when the main grid is lost they keep the island alive and share the load, without any change of control mode. This note describes the model as it runs today, reviews what the group of M. Liserre (Kiel University) and S. Bolognani (ETH Zürich) offer for this problem, and gives the first control version: P–f droop with a low-pass filter and a lead term (a virtual synchronous machine with an inertia constant of 1.25 s), Q–V droop, a virtual impedance in front of a voltage loop on the filter capacitor and of the existing current loop, current limitation and a time base distributed by PTP. Revision 01 replaces the virtual admittance of the first issue, which cannot hold an island of modules with LCL filters, analyses the small-signal stability of the whole discrete-time loop on stiff and weak grids and in the island, implements the control as the droop mode of the library block `grid_forming_dqctrl` and tests it on a light companion model: loss of the grid, load step, trip of a module, black start, clock errors and differences between the modules. It then treats the common mode of modules on one battery (common-mode chokes, PWM carriers aligned to a global PTP time, zero-sequence circulating currents), the earthing of an industrial battery energy storage system with the HF parasitic capacitances and earthing resistances now in the model, simulated for an unearthed, a high-resistance and a solidly earthed transformer star point, the residual current monitoring unit (RCMU) and the insulation monitor, and the step to three-level modules.



Revision history

Rev.	Date	Description	Authors
00	November 2026	First issue: state of the model, literature, control proposal V1 and test plan, common mode and PTP alignment, earthing with the HF parameters of the model, RCMU and IMD, three-level outlook.	Davide Bagnara
01	November 2026	Control V1 reworked: virtual impedance in front of a voltage loop on the filter capacitor instead of the virtual admittance; small-signal analysis of the discrete-time loop; V1 as the droop mode of the library block `grid_forming_dqctrl`; companion model `pgfm_droop` and results of the tests T0 to T5.	Davide Bagnara

Project Repository: grid-forming.git (parallel_of_GFMs)

Framework: MATLAB/Simulink/Simscape

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Nomenclature

Symbol	Meaning
AFE	active front end: three-phase PWM converter between a DC link and the AC network
BESS, PCS	battery energy storage system; power conversion system (the converters of the BESS)
GFM, GFL	grid-forming, grid-following
VSM	virtual synchronous machine
CM, ZSV, ZSCC	common mode; zero-sequence voltage; zero-sequence circulating current
PTP	precision time protocol, IEEE 1588
IT, TN, HRG	earthing systems: not earthed (or through a high impedance), solidly earthed, earthed through a high resistance
IMD	insulation monitoring device (IEC 61557-8)
RCMU, RCD	residual current monitoring unit; residual current device
S_b	base power of a module, 277.8 kVA (250 kW at $\cos\varphi = 0.9$)
u_bez, i_bez, Z_b	base voltage and current (peak values) and base impedance of a module, 563.4 V, 328.7 A, 1.714 Ω
ω_b	base angular frequency, $2\pi \cdot 50$ Hz
p, q; P_f, Q_f	active and reactive power of a module; the same after the low-pass filter (pu of S_b)
P^*, Q^*	set-points of a module from the plant controller
m_p, n_q	P–f and Q–V droop coefficients
m_d	gain of the lead term on the unfiltered power (damping of the swing modes)
E, U^*	amplitude of the internal voltage of a module; voltage set-point
$X_{\mathrm{c},k}$, γ_k	reactance of the connection of module k to the common bus; gain error of its voltage measurement
τ_p, τ_q	time constants of the power filters
H, D	inertia constant and damping of the equivalent virtual synchronous machine
R_v, X_v	virtual resistance and reactance
r_t, τ_hp	transient virtual resistance on the high-pass part of the output current; its time constant
u_c, i_s	voltage of the filter capacitor; converter current
u_g, i_o	voltage and current at the terminals of a module, after the grid-side inductor
u^*, i^*	references of the capacitor voltage and of the converter current
k_pu, k_iu; k_pi, k_ii	gains of the voltage PI; gains of the current PI
b	susceptance of the filter capacitor at 50 Hz in pu, $\omega_\mathrm{b} C_\mathrm{F} Z_\mathrm{b}$
i_max, i_thr	current limit; threshold of the virtual-impedance current limitation



Symbol	Meaning
L, L_0, C_F	filter of a module: positive- and zero-sequence inductance of the three-phase inductor, filter capacitance
L_1, L_2, X_2	LCL filter of the library block: converter-side and grid-side inductors; reactance of L_2
C_pole, C_ac	capacitance to earth of one battery pole, of one phase of the LV side
r_gnd, r_star	earthing resistor of one battery pole; resistance from the transformer star point to earth

1. Scope and summary

The model `pgfm_ctrl_analysis` of the folder `parallel_of_GFMs` of the repository `grid-forming` has three AFE modules of 250 kW in parallel on one lithium-ion battery and on a 690 V, 1 MVA grid emulator. Its purpose is the study of the *power sharing among grid-forming modules*: in normal operation the modules follow the set-points of the plant controller; if the main grid is lost, they must keep the voltage and the frequency of the island and share its load, ideally without detecting the event and without switching control mode. Two further topics belong to the same machine and are treated here because they decide what the power stage and the PWM must look like:

- the *common mode* of modules that share one battery and one AC bus: common-mode chokes and PWM carriers aligned to a global time distributed by PTP;
- the *earthing* of an industrial BESS: where the common-mode voltage of the bridges goes, which parasitic capacitances and earthing resistances to put in the model, how the insulation is monitored (IMD) and how the residual current is measured (RCMU).

The note proposes a first control version (V1) that can be implemented and simulated now, and a test plan to judge it. The main points are:

Control V1 (Section 5). Every module is a voltage source behind a virtual impedance: P–f droop with a low-pass filter on the measured power and a lead term ($m_\mathrm{p} = 2\,\%$, $\tau_\mathrm{p} = 50\,\mathrm{ms}$, the same as a virtual synchronous machine with $H = 1.25\,\mathrm{s}$ and $D = 50$), Q–V droop ($n_\mathrm{q} = 5\,\%$), a virtual impedance ($R_\mathrm{v} = 0.05$, $X_\mathrm{v} = 0.10\,\mathrm{pu}$, and a transient resistance) that gives the reference of a voltage loop on the filter capacitor, in front of the existing current loop, a current limit of 1.2 pu with a threshold virtual impedance, and the time base of the modules aligned by PTP. Rev. 01 replaces the virtual admittance of Rev. 00, which cannot hold an island of modules with LCL filters. There is no grid-loss detection and no mode change: on the loss of the grid the modules take the load at once and the frequency settles on the droop line ($-0.24\,\mathrm{Hz}$ in the test case). The small-signal analysis of the whole discrete-time loop gives a damping ratio of 0.45 to 0.47 to the swing modes and of at least 0.39 to the electromagnetic modes, from a stiff grid to a weak grid and in the island. The reactive sharing is limited by the accuracy of the voltage measurements more than by the impedances,



the active sharing by the clock errors if the time base is not aligned.

Implementation and tests (Sections 5.12 and 5.13). Done (route A): the control V1 is the droop mode of the library block `grid_forming_dqctrl`, and the companion model `pgfm_droop` (three modules, local load, grid breaker) passes the tests: loss of the grid, load step and trip of a module in the island without loss of synchronism, sharing of P and Q within 0.0001 pu, black start, peak converter current at most 1.04 pu. Next (route B): the validated control into `pgfm_ctrl_analysis` for the common-mode and earthing studies, which need its switching details.

Earthing (Section 7). Battery referenced to earth through a high impedance, LV side of the transformer not earthed (IT), insulation monitor, RCMU per PCS, inverter-duty transformer with an earthed screen. The simulation of the model with the new parameters shows why: with the star point not earthed the battery poles move by only 90 V peak to peak; with the star point solidly earthed the battery carries the whole common-mode voltage of the bridges, 1160 V peak to peak, and the earth current is seven times higher (4.4 A against 0.64 A rms).

2. The model today

2.1. Power stage

Figure 1 shows the model and Table 1 its data. Each module is the AFE part of the original `afe_abc_inv_psm` subsystem: a two-level bridge, a three-phase inductor and a star of filter capacitors with its star point floating. The three-phase inductor has $L = L_0 = 500\,\mu\text{H}$: its positive-sequence inductance is the filter inductance, its zero-sequence inductance makes it the common-mode choke of the module. There is no grid-side inductor in the modules: the capacitors of the three modules are on the same LV bus, and the grid-side inductance of the LCL filter is the one of the grid emulator (transformer 5 % and source 5 % on 1 MVA). The DC side of the three modules is one battery, the block `battery_1`, with the HF parasitics and the earthing resistors of Section 7.

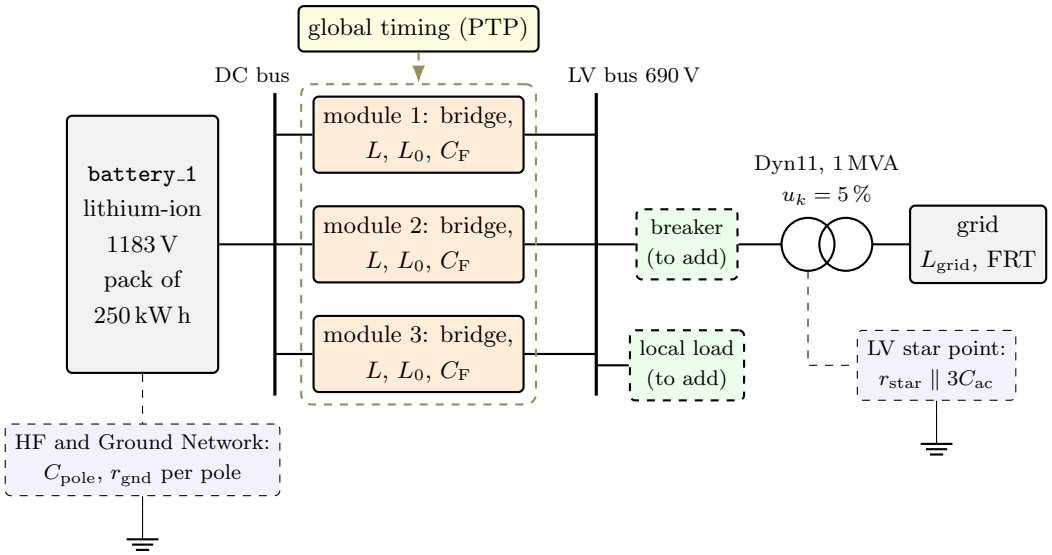


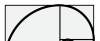
Figure 1: The model `pgfm_ctrl_analysis`: three modules on one battery and on the LV bus of the grid emulator. Blue: the earthing elements of Section 7; yellow: the clocks and the PWM carriers of the modules, aligned to the PTP time; green: the breaker and the local load that the tests of Section 5.13 need.

Table 1: Data of the model (`init_model.m`, November 2026).

Item	Value
modules	3, 250 kW each, $S_b = 277.8$ kVA, 690 V, 50 Hz
base values (per module)	$u_{bez} = 563.4$ V, $i_{bez} = 328.7$ A (peaks), $Z_b = 1.714\,\Omega$
three-phase inductor	$L = L_0 = 500\,\mu$H (0.092 pu), plus 5 µH leakage
filter capacitors	$C_F = 200\,\mu$F in star, star point floating (0.108 pu, 29.9 kvar)
DC link	7.2 mF per module, battery at 1183 V (SOC 0.85)
PWM and control	4 kHz, double update ($t_s = 125\,\mu$s), dead time 3 µs
solver step	$t_c = 1\,\mu$s (fixed step)
grid emulator	transformer Dyn11 690/690 V, 1 MVA, $u_k = 5\,\%$; source $L_{grid} = 75.8\,\mu$H (5 %)
battery	one pack of 250 kW at 1C, i.e. about 250 kW h
scenario	modules at $i_\xi^* = -0.85$ pu (734 kW into the battery), FRT test 25.4 of VDE-AR-N 4110 at 0.75 s for 250 ms, reactive current step at 1.335 s, 1.75 s in total
block budget	994 non-virtual blocks of the 1000 allowed by the Home license

2.2. Control and timing

The modules are grid-following today. Each one has a PLL, a current loop with resonant PI controllers ($k_p = 0.6$, $k_i = 35\,s^{-1}$), the fault ride-through logic of VDE-AR-N 4110 and a current reference set by the init ($i_\xi^* = -0.85$ pu, then the reactive-current steps after the fault). A zero-sequence term `u_cm_comp_modN` for the parallel operation is enabled by `param.afe.ref.en_parallel_mode` (0 for module 1, −1 for modules 2 and 3).



The block `global timing` gives each module its own clock with a gain error (fields `time_gain_module_N` of `param.afe.behav`: 1.0002, 1.0015, 0.9988, i.e. errors of 200 to 1500 ppm, deliberately large) and aligns it to the master time through a compensation loop and a transmission delay of 250 µs: this is the model of the time distribution by PTP. The PWM carriers of the modules run on the aligned time, with an optional phase shift (fields `pwm_phase_shift_modN`, zero today: carriers aligned).

2.3. What the model cannot do yet

- The modules cannot form the grid: they follow the PLL angle and a current reference (the control V1 is in the library block, not yet in this model).
- There is no breaker between the LV bus and the grid emulator and no local load, so the loss of the grid cannot be simulated (the companion model of Section 5.12 has both).
- The battery is sized for one module while the three modules draw 734 kW; the DC-link capacitors start without voltage (inrush of the battery in the first milliseconds).
- Every new function must be paid with blocks: 994 of 1000.

3. What the modules must do

The requirements below are the ones the first version is judged against; the numbers are proposals.

R1. **On the grid**: each module follows its set-points P^*, Q^* from the plant controller and, without anything added, gives inertia and a frequency response proportional to the frequency deviation (primary response).

R2. **Loss of the main grid**: the upstream breaker opens with a local load on the LV bus. No detection, no change of mode: the modules keep voltage and frequency (frequency within ± 1 Hz, positive-sequence voltage within ± 10 %) and share the load in proportion to their rating (error below 5 % of S_b for P, below 10 % for Q).

R3. **Robust sharing**: the sharing must survive different impedances between the modules and the bus (cables, tolerances), the errors of the voltage measurements and the clock errors of the modules.

R4. **N−1**: the trip of one module in the island must not stop the others.

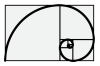
R5. **Current**: the converter current stays below 1.2 pu in every event, the module stays grid-forming while it limits.

R6. **Later**: frequency and voltage restoration in the island, resynchronization and reconnection to the grid, fault ride-through as a grid-forming unit (GB grid code GC0137 style).

4. Literature: Liserre, Bolognani and the base references

4.1. The group of M. Liserre (Kiel University)

The Chair of Power Electronics of Kiel University has worked on grid-forming converters in parallel and in mixed plants, mostly in the frame of virtual synchronous machines:



- the review of Rosso, Wang, Liserre, Lu and Engelken [1] classifies the grid-forming controls (droop, power synchronization, virtual synchronous machines, matching and oscillator-based controls), their inner loops (cascaded voltage and current loops, single loops, virtual impedance and admittance), their synchronization and their current limitation; it is the map used in this note to place the choice of Section 4.4;
- the robust stability of synchronverters (VSMs) operating in parallel [2], i.e. the case of this model, and of mixed plants with grid-forming and grid-following converters [3]: the second one matters as soon as the modules share a bus with grid-following units (PV inverters, or the present modules);
- a battery storage system in grid-forming mode used to damp sub-synchronous oscillations [4]: the same machine as here, seen from the grid;
- a current limitation strategy for grid-forming converters under symmetrical and asymmetrical faults that keeps the grid-forming behaviour before, during and after the fault, following the draft requirements of the British system operator and tested in power-hardware-in-the-loop [5]: the reference for the fault ride-through of version 2;
- the zero-sequence circulating current of converters in parallel on a common DC link and a common AC side without transformers: Zhu, Liserre, Chen and Wu show that the third-order zero-sequence voltage injected by the space vector modulation drives it and propose a modulation that removes its effect [6]; a further work analyses its spectrum and suppresses it with multi-resonant controllers [7]. This is exactly the situation of the three modules on one battery (Section 6).

4.2. S. Bolognani (ETH Zürich) and the group of F. Dörfler

S. Bolognani does not work on the inner control of a grid-forming converter but on the layer above it: how the set-points of many converters are computed so that the network is operated well. His results are the distributed control of the reactive power of the converters in a microgrid [8], the distributed feedback control of the reactive power for voltage regulation and loss minimization [9], and the online feedback optimization, in which an optimization algorithm runs as a controller driven by measurements rather than by a model of the network [10]. For this model they are the natural *tertiary* layer: the plant controller that computes P^* and Q^* (or the voltage set-points) of the modules from measurements, on top of the droop of each module.

In the same group, the grid-forming control with the strongest theoretical guarantees for converters in parallel is the dispatchable virtual oscillator control (dVOC) [11], which synchronizes the converters and shares the power with a single nonlinear law; the classic results on synchronization and power sharing of droop-controlled converters [12] and on the secondary frequency and voltage control by distributed averaging [13] are also from this school.

4.3. Base references

The droop control of converters in parallel goes back to Chandorkar, Divan and Adapa [14]; the design of the output impedance for the sharing of parallel UPS inverters and the hierarchical (primary, secondary, tertiary) control of microgrids are due to Guerrero et al. [15, 16]; the syn-



chronverter of Zhong and Weiss [17] and the power-synchronization control of Zhang, Harnefors and Nee [18] are the two other classic grid-forming laws; D'Arco and Suul [19] show that a droop with a low-pass filter on the power and a virtual synchronous machine are the same controller; Pogaku, Prodanović and Green [20] model and test the cascade of droops, virtual impedance, voltage loop on the filter capacitor and current loop in an inverter-based microgrid, and D'Arco, Suul and Fosso [21] build a virtual synchronous machine on the same cascade; Paquette and Divan [22] limit the current of grid-forming inverters with a virtual impedance; Golsorkhi, Lu and Guerrero [23] replace the frequency droop with a common time reference (GPS) and an angle droop. The textbook of Teodorescu, Liserre and Rodríguez [25] covers the converter-level part. The grid-forming block of the library and its control are described in [26, 27]; the effect of the PWM carriers of AFEs in parallel on the grid power quality in [28].

4.4. Choice for the first version

The first version uses the **droop with low-pass filter, equivalent to a virtual synchronous machine**, with a virtual impedance in front of a voltage loop on the filter capacitor and of the existing current loop:

- it is the structure most used in industry and in the grid codes for grid-forming units, and the one of the Kiel works on parallel VSMs [2, 1]; tuning it is tuning two physical numbers, inertia and droop;
- the sharing of active power in steady state is exact and needs no communication (it follows the droop coefficients); the sharing of reactive power is set by the droop and by the virtual impedance, which is chosen larger than the physical differences between the modules;
- it works in the same way on the grid and in the island: the loss of the grid is not an event for the control;
- the cascade keeps the existing current loop and its limitation; each module regulates the voltage of its own filter capacitor, and the virtual impedance with the grid-side inductor separates the modules on the bus, so that their voltage loops do not fight. Rev. 00 had chosen a virtual admittance in front of the current loop for fear of that fight; with LCL filters it leaves the voltage of an island to the resonance of the capacitors with the virtual inductors (Section 5.6);
- dVOC [11] is the comparison for version 2; the online feedback optimization of Bolognani [10] is the tertiary layer above the modules; with the PTP time in the plant the angle droop of [23] is an option that removes the frequency deviation.

5. Control V1: droop grid-forming modules with virtual impedance

Rev. 00 of this note proposed a virtual admittance in front of the existing current loop. The analysis of the discrete-time loop of the block and the averaged simulation of three modules showed that this structure cannot hold an island formed by modules with LCL filters (Section 5.6). Rev. 01 replaces it with the cascade of [20, 21]: the droops give the internal voltage, a virtual impedance gives the reference of the filter-capacitor voltage, a voltage loop on the capacitor gives

the reference of the existing current loop. This control is now the droop mode of the library block `grid_forming_dqctrl` (`p.ctrl.droop = 1`) and has been tested on the companion model `pgfm_droop` (Sections 5.12 and 5.13).

5.1. Structure

Figure 2 shows the control of one module. All quantities are in pu of the module base; the frames use the amplitude-invariant transformation, so that p and q are in pu of $S_\mathrm{b} = \frac{3}{2} u_\mathrm{bez} i_\mathrm{bez}$. u_c is the voltage of the filter capacitor, i_s the converter current, u_g and i_o the voltage and the current at the terminals of the module, after the grid-side inductor L_2. The measurements, the first-harmonic tracker, the current loop and the PWM are those of the block; the droop mode replaces its phase generator and its voltage loop.

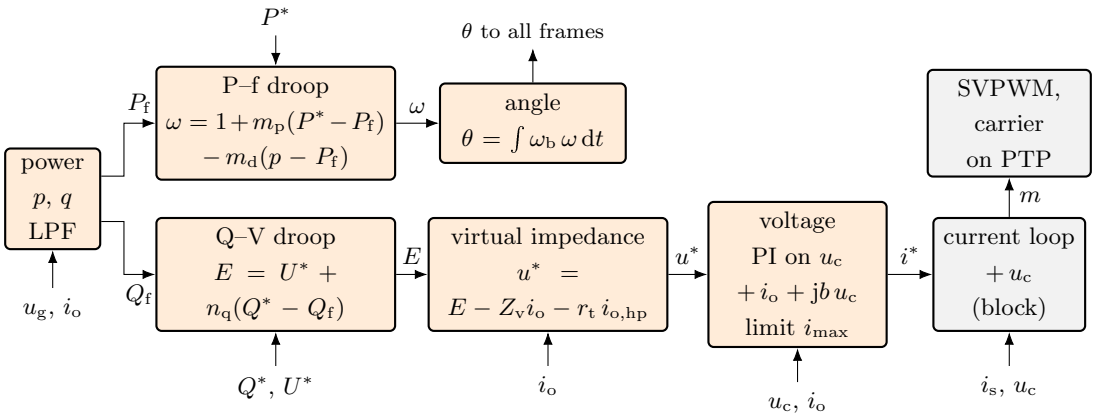


Figure 2: Control V1 of one module (Rev. 01). Orange: droop mode of the block; grey: parts of the block that the droop mode keeps. $Z_\mathrm{v} = R_\mathrm{v} + \mathrm{j}X_\mathrm{v}$, $i_\mathrm{o,hp} = i_\mathrm{o} - i_\mathrm{o,lp}$ the high-pass part of the output current, b the susceptance of the filter capacitor.

5.2. Power measurement

With u_g and i_o in the frame of the module (angle θ), sampled at the previous control step as everything in the block:

$$p = u_{gd} i_{od} + u_{gq} i_{oq}, \qquad q = u_{gq} i_{od} - u_{gd} i_{oq}, \qquad \tau_\mathrm{p}\, \dot{P}_\mathrm{f} = p - P_\mathrm{f}, \qquad \tau_\mathrm{q}\, \dot{Q}_\mathrm{f} = q - Q_\mathrm{f}. \tag{5.1}$$

These are the powers that the module delivers to the bus: the 0.108 pu of reactive power of its own filter capacitor stay inside the module and do not enter the droop. The filters at $\tau_\mathrm{p} = \tau_\mathrm{q} = 50\,\mathrm{ms}$ attenuate the 100 Hz ripple of an unbalanced load by a factor 31.

5.3. Active power: P–f droop with low-pass filter and lead term, a virtual synchronous machine

The frequency of the module and its angle are

$$\omega = 1 + \Delta\omega^* + m_\mathrm{p}\,(P^* - P_\mathrm{f}) - m_\mathrm{d}\,(p - P_\mathrm{f}), \qquad \dot{\theta} = \omega_\mathrm{b}\, \omega, \tag{5.2}$$



with P^* and the frequency offset $\Delta\omega^*$ from the plant controller. Without the lead term, eliminating P_f with (5.1) gives, for constant set-points,

$$\frac{\tau_\mathrm{p}}{m_\mathrm{p}}\,\dot\omega = P^* - p - \frac{1}{m_\mathrm{p}}\,(\omega - 1) \quad\Longleftrightarrow\quad 2H\,\dot\omega = P^* - p - D\,(\omega - 1), \tag{5.3}$$

the swing equation of a synchronous machine with $2H = \tau_\mathrm{p}/m_\mathrm{p}$ and $D = 1/m_\mathrm{p}$ [19]. With $m_\mathrm{p} = 0.02$ (1 Hz for 1 pu of power) and $\tau_\mathrm{p} = 50\,\mathrm{ms}$: $H = 1.25\,\mathrm{s}$, $D = 50$. The lead term on the unfiltered power ($m_\mathrm{d} = 0.01$) damps the swing modes without changing the droop, the steady state or the inertia seen by slow disturbances: without it the swing modes have a damping ratio of 0.15 (Table 2). In steady state all modules run at the same frequency, so $P_k^* - P_k = (\omega - 1)/m_\mathrm{p}$ is the same for all: with equal m_p and P^* the modules share the active power exactly, whatever their impedance.

5.4. Reactive power: Q–V droop

The amplitude of the internal voltage is

$$E = U^* + n_\mathrm{q}\,(Q^* - Q_\mathrm{f}), \qquad n_\mathrm{q} = 0.05, \tag{5.4}$$

U^* the voltage set-point (1 pu plus an offset from the plant controller). Unlike the frequency, the voltage is not the same at every module. The converter-side inductor L_1 is inside the loops and does not count; module k is its internal voltage behind the virtual impedance, its grid-side inductor ($X_2 = 0.015\,\mathrm{pu}$) and the reactance $X_{\mathrm{c},k}$ of its connection to the common bus (cable, busbar), and its reactive power is about

$$Q_k \simeq \frac{U^* + n_\mathrm{q} Q_k^* - (1 + \gamma_k)\,U - R_\mathrm{v} P_k}{X_\mathrm{v} + X_2 + X_{\mathrm{c},k} + n_\mathrm{q}}, \tag{5.5}$$

U the bus voltage and γ_k the gain error of the voltage measurement of the module. With $X_\mathrm{v} = 0.10\,\mathrm{pu}$ and $n_\mathrm{q} = 0.05$ the denominator is 0.165 pu: a connection 0.01 pu longer (55 µH) takes 6 % of the reactive power of the module, a gain error of 1 % moves its reactive power by $\gamma U/0.165 = 0.056\,\mathrm{pu}$ of S_b: in the island the accuracy of the voltage measurement, not the impedances, sets the reactive sharing (test T4). The term $R_\mathrm{v} P_k$ couples the active power into the reactive one: a module that delivers ΔP more than the others takes $\Delta Q \simeq -R_\mathrm{v}\Delta P/0.165 = -0.3\,\Delta P$ (test T1); with equal set-points it has no effect.

5.5. Virtual impedance, voltage loop and current loop

The reference of the capacitor voltage is the internal voltage $e = (E, 0)$ in the frame θ minus the drop on a virtual impedance:

$$u^* = E - (R_\mathrm{v} + \mathrm{j}X_\mathrm{v})\,i_\mathrm{o} - r_\mathrm{t}\,(i_\mathrm{o} - i_\mathrm{o,lp}), \qquad \tau_\mathrm{hp}\,\dot i_\mathrm{o,lp} = i_\mathrm{o} - i_\mathrm{o,lp}. \tag{5.6}$$

$X_\mathrm{v} = 0.10\,\mathrm{pu}$ (0.55 mH) dominates the physical differences between the modules and decouples P from Q; $R_\mathrm{v} = 0.05\,\mathrm{pu}$ (86 mΩ) and the transient resistance $r_\mathrm{t} = 0.15\,\mathrm{pu}$ on the high-pass part



of i_o ($\tau_{hp} = 5\,\text{ms}$, no effect in steady state) damp the electromagnetic modes. The impedance is quasi-static, the product $jX_v i_o$ in the frame without the derivative of i_o, so that no measured current is differentiated.

A vector PI in the frame regulates the capacitor voltage, with the feedforward of the output current and of the capacitor current at the fundamental:

$$i^* = k_{pu}\,(u^* - u_c) + k_{iu}\int (u^* - u_c)\,dt + i_o + j\,b\,u_c, \qquad |i^*| \le i_{max}, \tag{5.7}$$

$b = \omega_b C_F Z_b = 0.108\,\text{pu}$, $k_{pu} = 0.8$, $k_{iu} = 80\,\text{s}^{-1}$; the limiter clips on a circle and holds the integrators while it clips. The current loop of the block follows i^* and receives u_c as feedforward of the modulation index, so that its PI only corrects the drop on L_1:

$$m = k_{pi}\,(i^* - i_s) + k_{ii}\int (i^* - i_s)\,dt + \frac{u_c}{k_m}, \qquad k_m = \frac{u_{dc}}{\sqrt{3}\,u_{bez}}, \tag{5.8}$$

with the gains of the block ($k_{pi} = 0.75$, $k_{ii} = 35\,\text{s}^{-1}$) and u_{dc} the acquired DC-link voltage.

5.6. Why a cascade and not the virtual admittance of Rev. 00

Virtual admittance with LCL modules. In Rev. 00 the internal voltage drove a virtual inductor towards the measured voltage and its current was the reference of the current loop: each module was a current source behind a virtual admittance. With LCL filters the filter capacitors of the modules are on the bus, and in the island nothing else sets its voltage: the bus is a resonant circuit of the capacitors and the virtual inductors, damped by R_v alone, and the delays of the voltage measurement and of the current loop make it unstable. In the averaged model of the three modules (ideal current loop with a lag of 0.4 ms, voltage measurement filtered at 2 ms) the structure holds the operation on the grid, but after the loss of the grid the bus voltage oscillates between 0.5 and 3.9 pu within 0.4 s and the currents run into the limit; with faster measurements (0.1 ms) it is unstable already on the grid. A grid-forming module with an LCL filter must regulate the voltage of its own capacitor.

Feedforward of the output current. Without the feedforward of i_o in (5.7) the voltage PI alone must supply the whole output current, its integrator becomes the slowest element of the chain and the swing mode is unstable on every grid ($\zeta = -0.28$ at 2.4 Hz, Table 2): on a stiff grid the module slips. With the feedforward the PI only corrects the errors, the capacitor voltage follows u^* within a few milliseconds and the module is the voltage source that the droop assumes. The voltage PI of the block ($k_{pu} = 0.5$, $k_{iu} = 18\,\text{s}^{-1}$) is stable with the feedforward but leaves the swing modes at $\zeta = 0.23$; $k_{pu} = 0.8$, $k_{iu} = 80\,\text{s}^{-1}$ give 0.45.

5.7. Current limitation

Two stages. Above $|i_o| = i_{thr} = 1.0\,\text{pu}$ a threshold virtual impedance [22] adds $\Delta X_v = k_X(|i_o| - i_{thr})$ with $k_X = 1$ and $\Delta R_v = \Delta X_v/5$, so that the module stays a voltage source behind a larger impedance; the reference of the current loop is clipped on a circle of radius $i_{max} = 1.2\,\text{pu}$ with



the integrators of the voltage PI held (anti-windup). While the module limits, the measured power is lower than the one it would deliver and the droop raises its frequency: in version 1 this is accepted (it is what makes an overloaded module shed load to the others); the fault ride-through with a frozen or bounded frequency is version 2 [5].

5.8. Start, synchronization, black start and loss of the grid

Until t_{sync} (0.05 s in the example parameters of the block, 0.1 s in the companion model) the module follows the terminal voltage: its angle is that of the first-harmonic tracker of u_{g}, turned onto the sampled capacitor voltage with a time constant of 5 ms, its current reference is zero and the power filters and the low-pass filter of i_{o} run. At t_{sync} the droop starts from that angle and that operating point without a step, and the set-points P^*, Q^* ramp from zero in $t_{\mathrm{ramp}} = 0.1\,\mathrm{s}$. The turn matters: the output of the tracker leads the samples of the previous trigger by about 0.07 rad, and a voltage loop started in its frame pulls u_{c} forward by as much, a kick of 0.36 pu of active power on the grid that the droop then takes back with a dip of 0.13 Hz; with the turn the kick is 0.04 pu and the frequency does not dip. If the bus is dead at t_{sync} (tracker output below 0.5 pu) the module makes a black start: its angle runs free from θ_0 and E ramps from zero in t_{ramp}; modules that black-start together synchronize through the droop, whatever their initial angles. On the loss of the grid nothing changes in the control: the modules, which were voltage sources, take the power that the grid supplied; the frequency moves with the inertia $2H$ and settles at the droop value. The plant controller (version 2) restores the frequency and the voltage, synchronizes the island to the grid when it comes back and closes the breaker.

5.9. Small-signal analysis

Rev. 00 checked the power loops with the swing equation (5.3) of a module seen as an ideal voltage source behind X_{v}. Rev. 01 linearizes the whole discrete-time loop of the block: three modules with their LCL filters, the bus with the local load (R in parallel with L and its series resistance), the grid behind $R+L$, and the control of each module as in the block at $t_{\mathrm{s}} = 125\,\mu\mathrm{s}$, with the one-step delay of the samples, the filters, the droops, the virtual impedance, the two PIs with their feedforwards and the modulation. The map $z_{k+1} = F(z_k)$ is written in a frame rotating at 50 Hz; its equilibrium is found by Newton's method, its Jacobian by central differences, and each eigenvalue λ gives $s = \ln\lambda / t_{\mathrm{s}}$. Table 2 gives the least-damped mode in two bands: the swing modes of the droops (below 20 Hz) and the electromagnetic modes of the voltage loops with the filters and the network (20 to 500 Hz), for a stiff grid ($X_{\mathrm{grid}} = 0.01$ pu), the grid of the model (10 % on 1 MVA, 0.028 pu), a weak grid ($X_{\mathrm{grid}} = 0.11$ pu, short-circuit ratio 3 for the three modules) and the island; the load is 600 kW + 200 kvar, $P^* = 0.4$ pu.



Table 2: Least-damped modes of the discrete-time loop of three modules: damping ratio ζ and frequency, swing modes (below 20 Hz) and electromagnetic modes (20 to 500 Hz).

Design	Modes	stiff grid	grid of the model	weak grid, SCR 3	island
V1, 690 V module	swing	0.45, 5.6 Hz	0.46, 5.6 Hz	0.47, 5.5 Hz	0.47, 5.3 Hz
	electromagnetic	0.61, 75 Hz	0.56, 68 Hz	0.39, 56 Hz	0.64, 80 Hz
V1, 400 V example of the block	swing	0.43, 5.4 Hz	0.44, 5.3 Hz	0.45, 5.2 Hz	0.46, 5.1 Hz
	electromagnetic	0.41, 61 Hz	0.39, 59 Hz	0.30, 55 Hz	0.43, 62 Hz
without r_t	electromagnetic	0.07, 86 Hz	0.07, 86 Hz	0.07, 86 Hz	0.07, 86 Hz
without m_d	swing	0.14, 4.9 Hz	0.14, 4.8 Hz	0.15, 4.8 Hz	0.16, 4.7 Hz
voltage PI of the block (0.5, 18 s^{-1})	swing	0.23, 5.1 Hz	0.23, 5.1 Hz	0.24, 5.0 Hz	0.25, 4.9 Hz
without feedforward of i_o	swing	−0.29, 2.4 Hz	−0.28, 2.4 Hz	−0.27, 2.4 Hz	−0.25, 2.3 Hz

With the parameters of Table 3 the least-damped mode has $\zeta = 0.39$ (weak grid, 56 Hz) for the 690 V module and 0.30 for the 400 V example of the block (330 µH, 360 µF, 80 µH), whose parameter file carries the same values. A light island load (0.3 pu) or $P^* = 1$ pu change the damping by less than 0.02. The electromagnetic modes between 55 and 86 Hz are those of the voltage loops with the filter and the network; the transient resistance r_t takes them from 0.07 to 0.4–0.6. The swing modes at about 5.5 Hz owe their damping to the lead term and to the speed of the voltage loop. The expected values of Rev. 00 for the tests ($\zeta = 0.38$ on the grid and 0.28 in the island, from the swing equation alone) are replaced by these.

5.10. Clock errors, PTP and power sharing

The droop (5.2) is computed on the time of the module. If its clock runs fast by ε, the frequency that the module believes to impose is wrong by ε, and in steady state, where all modules have the same real frequency,

$$P_k = P_k^* - \frac{\omega - 1 - \varepsilon_k}{m_\mathrm{p}} \quad \Rightarrow \quad P_j - P_k = \frac{\varepsilon_j - \varepsilon_k}{m_\mathrm{p}}. \tag{5.9}$$

With the gains of the model, modules 2 and 3 are 2700 ppm apart: without the time alignment they share with an error of 13.5 % of S_b (test T0: 0.1345 pu); a 50 ppm crystal gives 0.25 % per module. The alignment of the clocks by PTP (the `global timing` of the model, the alignment loop of the block) removes this error (T0: 0.0003 pu) and is the condition for the angle droop of version 2. The same time base aligns the PWM carriers (Section 6).



5.11. Parameters of V1

Table 3: Parameters of the control V1 (pu of the module base) and fields of `p.ctrl` of the block.

Parameter	Value	Field	Note
P–f droop m_{p}	0.02	`droop_mp`	1 Hz per pu; $D = 50$
power filters τ_{p}, τ_{q}	50 ms	`droop_tau_p`, `_q`	$H = 1.25$ s
lead term m_{d}	0.01	`droop_md`	damping of the swing modes
Q–V droop n_{q}	0.05	`droop_nq`	
virtual impedance R_{v}, X_{v}	0.05, 0.10	`droop_rv`, `_xv`	86 mΩ, 0.55 mH
transient resistance r_{t}, τ_{hp}	0.15, 5 ms	`droop_rt`, `_tau_hp`	damping of the electromagnetic modes, no effect in steady state
threshold virtual impedance	$i_{\mathrm{thr}} = 1.0$, $k_X = 1$, $X/R = 5$	`droop_i_thr`, `_kx`, `_x_r`	[22]
current limit $i_{\max}$	1.2	`droop_i_max`	clip of i^*, integrators held
voltage PI k_{pu}, k_{iu}	0.8, 80 s^{-1}	`droop_kp_u`, `_ki_u`	feedforward of i_{o} and $\mathrm{j}bu_{\mathrm{c}}$
current loop k_{pi}, k_{ii}	0.75, 35 s^{-1}	`kp_i`, `ki_i`	of the block, feedforward of u_{c}
start t_{sync}, t_{ramp}	0.05 (0.1), 0.1 s	`droop_t_sync`, `_t_ramp`	0.1 in the companion model
set-points P^*, Q^*, U^*	0.4, 0, 1	`droop_p_ref`, `_q_ref`, `ug_ref`	offsets from the input `sync`: [–; ΔP^*; ΔQ^*; ΔU^*; $\Delta\omega^*$]
time base	PTP	`k_time`, `kp_align`, `ki_align`	clocks and carriers aligned

5.12. How V1 is put in the models

Route A (done in Rev. 01): library block and a light model. The control V1 is the droop mode of the MATLAB Function of the library block `grid_forming_dqctrl` [26] (`p.ctrl.droop = 1`, fields `droop_*`; `ctrl_log` from 43 to 55 signals, 44 to 55 for the droop; the input `sync` carries the set-point offsets; the ports do not change and the existing models are not affected). The block with the delta capacitors, `grid_forming_dqctrl_delta`, links the same code. The companion model `pgfm_droop.slx` in `parallel_of_GFMs/power_sharing` has three such blocks with the switching bridge, each on an ideal DC source, a switch in the output of each module (trip, with an RC snubber across each pole that takes the current of L_2 when it opens, as the arc of a contactor does until the current zero) and an optional cable, the 690 V bus with the local load and a switched load step, and the grid (10 % on 1 MVA) behind a breaker: 277 blocks, Simscape Foundation only, built by `build_pgfm_droop.m`, parameters and cases in `init_pgfm_droop.m`, tests in `pgfm_droop_tests.m`. It simulates about 0.7 s per minute on a desktop PC. The battery with its earthing network and the grid emulator are not in it: common mode and earthing stay with route B.

Route B: the present model. In `pgfm_ctrl_analysis` the PLL and the reference chain of each module (about 60 non-virtual blocks) are replaced by one MATLAB Function with the control V1, in front of the existing current loops; the blocks saved pay for the breaker and the load. It keeps the switching details, the PTP model and the common-mode paths, but it changes the control of the modules that the model was built around, and the license limit stays tight.



The two routes are complementary: A to design and validate the control, B afterwards to study common mode, earthing and PTP with the validated control.

5.13. Tests on the companion model

Module of 250 kW at 690 V (S_b = 277.8 kVA), L_1 = 500 µH, C_F = 200 µF, L_2 = 80 µH, u_{dc} = 1070 V, PWM at 4 kHz with double update, solver step 1 µs; local load 600 kW + 200 kvar at 690 V (constant impedance), P^* = 0.4 pu, Q^* = 0. On the grid the breaker closes at 1 ms and the modules synchronize until 0.1 s. Table 4 gives the events and the results; the frequency is that of the phase generators, the voltage $|u|$ the magnitude of the bus voltage averaged over 1 ms, the rate of change of frequency the largest derivative of the frequency averaged over 10 ms, from 10 to 200 ms after the event, the settling time the time to enter a band of ±0.01 pu around the final P_f.

Table 4: Tests of the control V1 on `pgfm_droop` (powers in pu of S_b, per module).

Test	Event	Result
T0	island (black start); clocks of the modules 1.0002, 1.0015, 0.9988; time alignment off, then on	off: P = 0.639, 0.704, 0.569 pu, spread 0.1345 pu as (5.9); on: spread 0.0003 pu
T1	on the grid: P^* of module 1 from 0.4 to 0.8 pu at 0.6 s	P_f of module 1 settles in 0.17 s, overshoot 5 %; its frequency steps by $m_p\,\Delta P^*$ = 0.4 Hz and comes back to 50 Hz; $P = P^*$ within 0.0001 pu; Q of module 1 −0.14 pu (5.5: −0.12, the rest is the angle across X_v); peak current 1.04 pu
T2	loss of the grid at 1.0 s, 600 kW + 200 kvar on the bus	50.000 to 49.762 Hz (nadir 49.758), 4.5 Hz/s; $|u|$ 0.978 to 0.926 pu, minimum 0.82 pu; P 0.40 to 0.638 pu and Q to 0.204 pu each, settled in 0.15 s, overshoot 0.1 %; peak current 0.72 pu
T3	island: load step of +150 kW at 1.5 s (after T2)	to 49.623 Hz (nadir 49.618), 2.7 Hz/s; $|u|$ to 0.917 pu, minimum 0.85 pu; P to 0.777 pu and Q 0.201 pu each, settled in 0.13 s; spread of P and Q below 0.0001 pu; peak current 0.86 pu
T4	island (black start): module 3 with L_1 + 20 %; with a cable of 0.01 pu; with U^* 1 % low (a gain error of 1 %)	spread of Q: 0.002 pu (no effect); 0.011 pu (5.5: 0.012); 0.058 pu (5.5: 0.056; Rev. 00 expected 0.045). P shared within 0.0001 pu in the three cases
T5	450 kW + 150 kvar: loss of the grid at 1.0 s, trip of module 3 at 1.5 s	loss of the grid: 49.903 Hz, P 0.40 to 0.497 pu; trip: modules 1 and 2 to P = 0.704 pu, Q = 0.224 pu at 49.696 Hz (nadir 49.690, 3.7 Hz/s), minimum voltage 0.84 pu, settled in 0.14 s, peak current 0.79 pu; module 3 runs on at no load
black start	no grid, 600 kW + 200 kvar, start at 0.1 s	bus voltage at 90 % in 0.09 s, overshoot 1.4 %; 49.762 Hz, P = 0.638 pu, Q = 0.204 pu each, peak current 0.71 pu

All cases pass the checks of `pgfm_droop_tests.m`: equal sharing in the island (spread below 0.01 pu of P and 0.02 pu of Q, except where the test introduces a difference), $P = P^*$ on the grid, frequency on the droop line, damped power (peak to peak of P_f below 0.02 pu at the end) and converter current below i_{max}. The frequency after the loss of the grid is −0.24 Hz and not −0.32 Hz as estimated in Rev. 00: the load is a constant impedance and the island settles at

0.926 pu, so the load takes 14 % less power; for the same reason the two modules left in T5 take 0.704 pu and not the 0.81 pu of a constant-power load. That voltage, 7 % below the nominal one with the modules at 0.64 pu, is the drop of the Q–V droop and of the virtual impedance; restoring it, with the frequency, is the task of the secondary control of version 2. In the island the filtered powers follow the load with the lag of their filters (settled in about $3\tau_\mathrm{p} = 0.15$ s, overshoot at most 1 %): the modules are voltage sources and take the load at once, and events that hit the three equal modules alike do not excite the swing modes between them. T1 excites the swing mode of module 1 against the grid: 5 % of overshoot of P_f. Figures 3 and 4 show tests T2–T3 and T5.

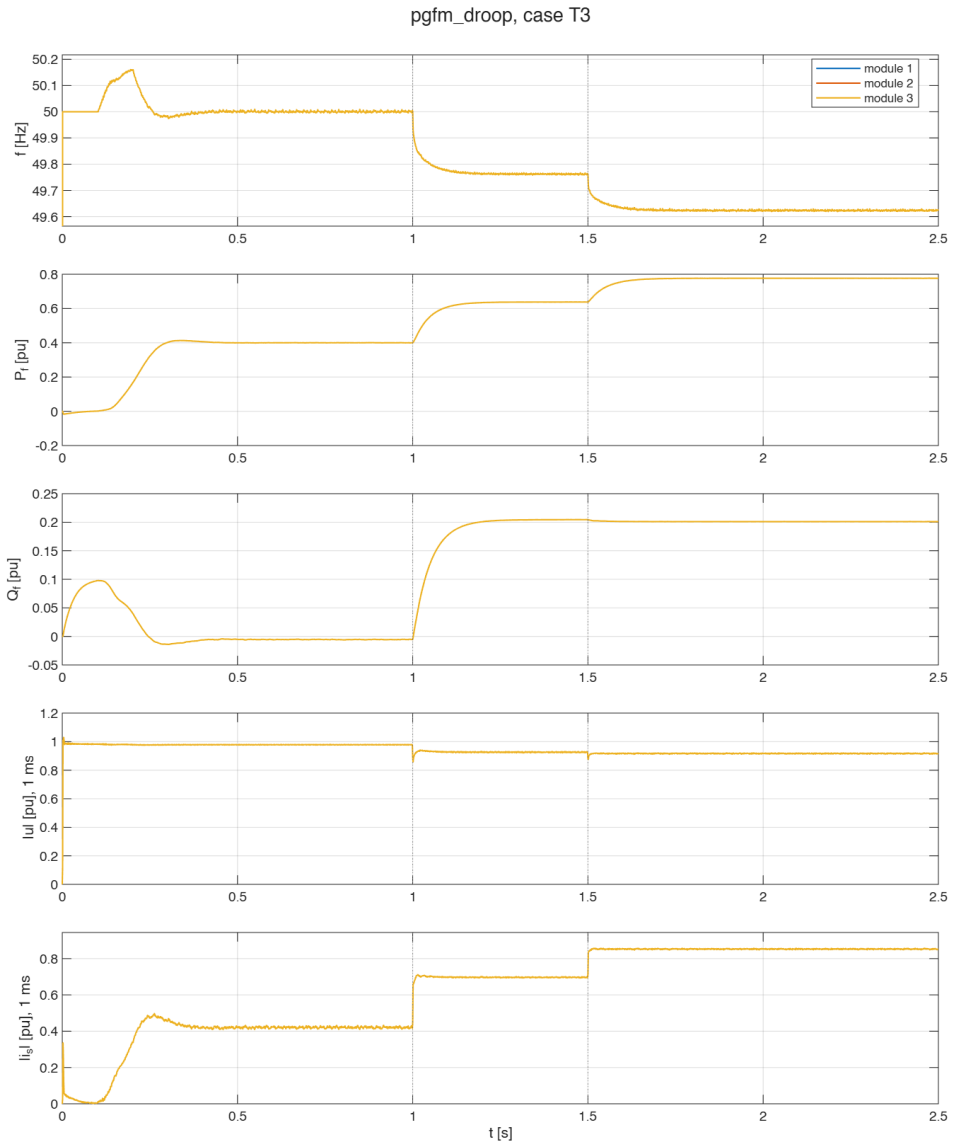


Figure 3: Tests T2 and T3 on `pgfm_droop`: start on the grid, loss of the grid at 1.0 s, load step of +150 kW at 1.5 s. From the top: frequency of the phase generators, filtered active and reactive power, bus voltage and converter current (magnitudes, 1 ms averages); the three modules overlap. Until 0.1 s the modules follow the grid with zero converter current and Q_f is the reactive power of their filter capacitors (0.10 pu); then P^* ramps to 0.4 pu in 0.1 s and the frequency rises to 50.16 Hz while the angles advance.

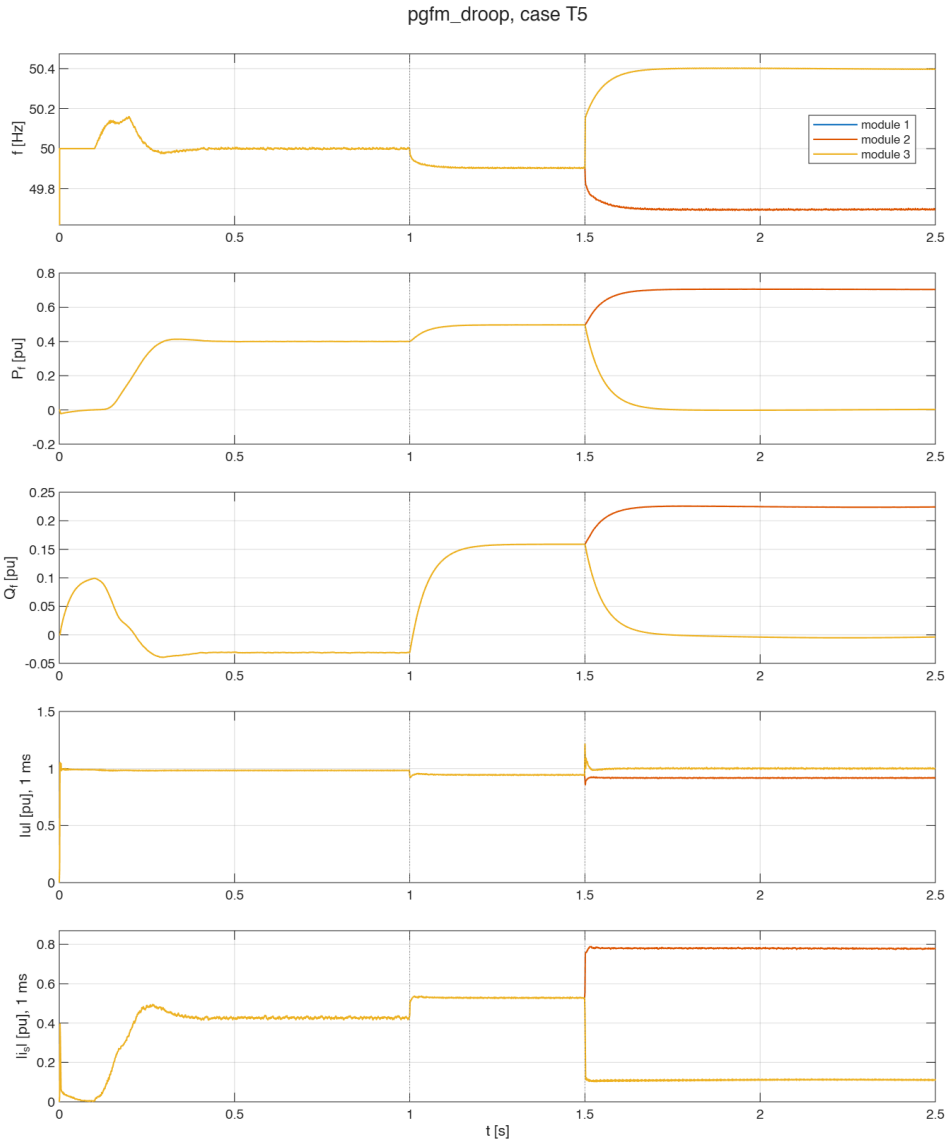


Figure 4: Test T5 on `pgfm_droop`: 450 kW + 150 kvar, loss of the grid at 1.0 s, trip of module 3 at 1.5 s; after the trip module 3 runs alone at no load (50.4 Hz). The spike of $|u|$ at the trip, 1.22 pu (above 1.05 pu for 7 ms), is the voltage at the terminals of module 3, on its side of the open switch, which loses its load at once; the bus dips to 0.84 pu.

5.14. Version 2 and later

Secondary control (frequency and voltage restoration in the island by the plant controller, or by distributed averaging [13]); resynchronization and reconnection; fault ride-through as grid-forming units with the current limitation of [5] (test T6: FRT test 25.4 on the grid, current below 1.2 pu, recovery, synchronism); angle droop on the PTP time [23] with the frequency droop as fallback if the time is lost; dVOC as comparison [11]; the tertiary layer by online feedback optimization [10].



6. Common mode of modules on one battery

6.1. Where the common-mode voltage comes from

The common-mode voltage of a two-level bridge, measured from the midpoint of its DC link, is the average of the three leg voltages, $v_{\mathrm{cm}} = (v_{a0} + v_{b0} + v_{c0})/3 \in \{\pm u_{\mathrm{dc}}/2, \pm u_{\mathrm{dc}}/6\}$: at every commutation it jumps by $u_{\mathrm{dc}}/3$, 394 V at 1183 V. Its low-frequency part is the zero-sequence voltage that the modulator adds to the references: the min–max injection of the space vector modulation is a triangle at three times the fundamental with a peak of a quarter of the phase amplitude, i.e. about 114 V at 150 Hz at full modulation. The simulation of Section 7.3 gives 113 V at 150 Hz (in the IT case 10 V on the battery and 103 V on the LV star point).

Each module pushes its common-mode voltage through its common-mode inductance. For the three-phase inductor of the model (Appendix A), the zero-sequence inductance L_0 is per phase; the total common-mode current $i_{\mathrm{cm}} = i_a + i_b + i_c$ of a module sees $L_0/3 = 168\,\mu\mathrm{H}$ (leakage included).

6.2. Circulating current between modules

On one battery and one AC bus without transformers the modules form loops in which only the difference of their common-mode voltages acts (Figure 5): the zero-sequence circulating current

$$\frac{2L_0}{3}\,\frac{\mathrm{d}i_{\mathrm{zs}}}{\mathrm{d}t} = v_{\mathrm{cm},1} - v_{\mathrm{cm},2} \tag{6.1}$$

does not reach the grid and is not seen by a current loop in the dq frame, but it loads the semiconductors and the chokes. Two sources matter:

- **different references.** The 150 Hz zero-sequence voltage depends on the reference of each module [6]: 10 % of difference in the modulation (11 V at 150 Hz) drives 36 A through the 337 µH of the loop, i.e. 12 A per phase. With equal sharing (control V1) the references are close and this term is small; in transients, or with modules at different set-points as in the present scenario, it is not;
- **misaligned carriers.** With the carriers shifted, the common-mode pulses of the modules no longer coincide: half a carrier period of shift gives a ripple of about 70 A peak to peak at the switching frequency. Interleaved carriers reduce the ripple seen by the grid [28] but maximize this current: in parallel on one DC link, aligned carriers are the default.

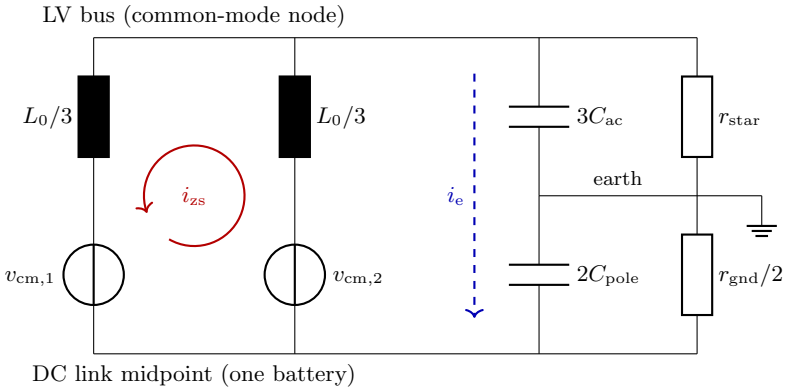


Figure 5: Common-mode paths of modules on one battery (lumped, one source per module referred to the DC midpoint). Red: circulating current between the modules; blue: current to earth through the parasitic capacitances (Section 7).

6.3. PWM carriers aligned to the PTP time

The alignment needs a common time in every module. IEEE 1588 (PTP) on a local network with hardware time stamps and PTP-aware switches keeps the clocks within some tens to a few hundred nanoseconds. With 100 ns of residual shift each common-mode edge leaves 394 V × 100 ns ≃ 39 µV s on 337 µH, about 0.1 A: negligible. Without a common time, even good crystals drift apart: 1500 ppm (module 2 of the model) loses half a carrier period in 83 ms, 50 ppm in 2.5 s. The same time base gives the clocks of the droop (Section 5.10) and the time stamps of the measurements.

6.4. Measures in the model and next steps

- keep the carriers aligned (`pwm_phase_shift_modN` = 0) and the alignment loop of `global timing` on; run once with the alignment off to see the drift and the circulating current;
- the same modulator and the same zero-sequence injection in all modules; the term `u_cm_comp_modN` must be checked against (6.1): different values between the modules create a 150 Hz circulating current;
- log $i_{cm,k} = i_a + i_b + i_c$ of every module (a sum block per module, no new physical block);
- version 2: a loop per module on $i_{cm,k}$ acting on its zero-sequence voltage, PI plus resonant terms at 150 and 450 Hz [7], or the modulation of [6].

7. Earthing of an industrial BESS and HF parasitics

7.1. Earthing schemes

The common-mode voltage of the bridges has to close somewhere: through the parasitic capacitances to earth of the DC side (battery, cables) and of the AC side (transformer winding, cables, AC nodes of the modules). Where it appears depends on which side is held to earth:

- **LV star point solidly earthed (TN)**, battery floating: the AC side is fixed to earth, so the battery carries the whole common-mode voltage of the bridges with respect to earth.



Every cell, its insulation towards the racks and the cooling plates, and the cables see steps of $u_{\mathrm{dc}}/3$ at the switching frequency; the earth current through the capacitance of the battery is the largest of all cases. This was the scheme of the model before November 2026.

- **LV side not earthed (IT), battery referenced to earth through a high impedance**: the common-mode voltage divides on the capacitances of the two sides; the battery, much larger than the AC side, stays almost still and the LV winding carries the common mode. It is the scheme of the industrial BESS with a power conversion system and an LV/MV transformer: the LV winding and everything galvanically connected to it (bridges, DC links, battery) form one IT system, with an insulation monitor (first fault signalled, not tripped) and residual current monitoring.
- **high-resistance earthing (HRG) of the LV star**: a resistor limits the first-fault current (here 1.7 A) and makes it detectable; below the corner frequency $1/(2\pi r_{\mathrm{star}}C)$ (2.1 kHz here) the star point is held to earth, so the 150 Hz common mode goes back onto the battery, and the resistor carries a permanent common-mode current.

The transformer between the LV system and the MV network should be an inverter-duty unit with an earthed electrostatic screen between the windings: the common-mode current of the LV system then closes locally and is not coupled into the MV network, whose neutral treatment is the one of the network operator.

7.2. Parasitic capacitances and earthing resistances in the model

Table 5 gives the values now in `init_model.m` (`param.gnd`); Figure 5 shows where they are. The battery side is the network drawn in `battery_1/HF and Ground Network`: from each pole a capacitor with its series resistance and an earthing resistor to earth. The AC side has the three phase capacitances lumped at the star point of the LV winding of the emulator transformer (`grid/grid_emulator/3ph_Dyg_ssc`): the block `C_lv_earth` ($3C_{\mathrm{ac}}$ with $r = r_{\mathrm{ac,hf}}/3$) in parallel with `R_star`. At the frequencies of interest the star point and the phases have the same common-mode voltage, and the lumped form costs two blocks less.

Table 5: HF parasitics and earthing in `param.gnd`.

Field	Value	Origin
`c_batt_pole`	150 nF	cells to racks and cooling plates (about 1 nF per prismatic cell of 280 A h through its insulation film), DC cables and busbars: about 0.3 µF for a 250 kW h string, half per pole; it scales with the strings (about 0.6 µF per pole for 1 MWh)
`r_batt_hf`	10 Ω	series resistance of that path: earth conductors, skin effect, losses of the common-mode cores; it damps the common-mode resonance
`r_gnd_pole`	100 kΩ	symmetric high-resistance reference of the poles: first-fault current $u_{\mathrm{dc}}/r = 12$ mA, 7 W of losses; with an IMD, 1 MΩ or more (the IMD sees $r/2$)
`c_ac_phase`	10 nF	LV winding to core and screen, LV cables, AC nodes of the IGBT modules to the heatsinks, per phase
`r_ac_hf`	10 Ω	series resistance of that path, per phase
`r_star`	1 GΩ	IT (not earthed); 230 Ω for high-resistance earthing (1.7 A); 0.1 Ω for TN



With these values the capacitive divider gives the battery $C_{\mathrm{ac,tot}}/(C_{\mathrm{ac,tot}}+C_{\mathrm{dc,tot}}) = 30/330 = 9.1\,\%$ of the common-mode voltage in the IT case. The common-mode loop (the chokes of the three modules in parallel, about 56 µH, the zero-sequence leakage of the transformer and the 27 nF of the two capacitances in series) resonates near 100 kHz; with $t_{\mathrm{c}} = 1$ µs the model resolves it with about ten steps per period, so the amplitude of that ringing is indicative only, while the 150 Hz and the switching-frequency components are reliable.

7.3. Simulation of the three schemes

The script `parallel_of_GFMs/earthing_cases.m` runs the model to 0.3 s for the three values of r_{star}, with the Simscape log of the earth paths, and evaluates the last three grid periods (Table 6, Figure 6); the operating point is the one of the scenario, 734 kW into the battery at 1183 V.

Table 6: Earthing schemes: voltages to earth and current to earth, last three grid periods of a run to 0.3 s (734 kW into the battery at 1183 V). 150 Hz: amplitude of that component; bands: rms value of the spectrum of the window in the band.

	IT	**HRG**	**TN**
r_{star}	1 GΩ	230 Ω	0.1 Ω
DC+ to earth, peak to peak [V]	91	410	1158
battery (DC midpoint) to earth, 150 Hz [V]	10	112	113
LV star point to earth, peak to peak [V]	830	607	3
LV star point to earth, 150 Hz [V]	102	7	0
current to earth, rms [A]	0.64	0.71	4.41
current to earth, peak [A]	2.6	2.7	15.7
band 0–2 kHz, rms [mA]	3	24	24
band 2–20 kHz, rms [A]	0.20	0.46	3.90
band 20–500 kHz, rms [A]	0.60	0.54	2.06
current in r_{star}, rms [A]	0	0.46	4.41
losses in r_{star} [W]	0	49	2

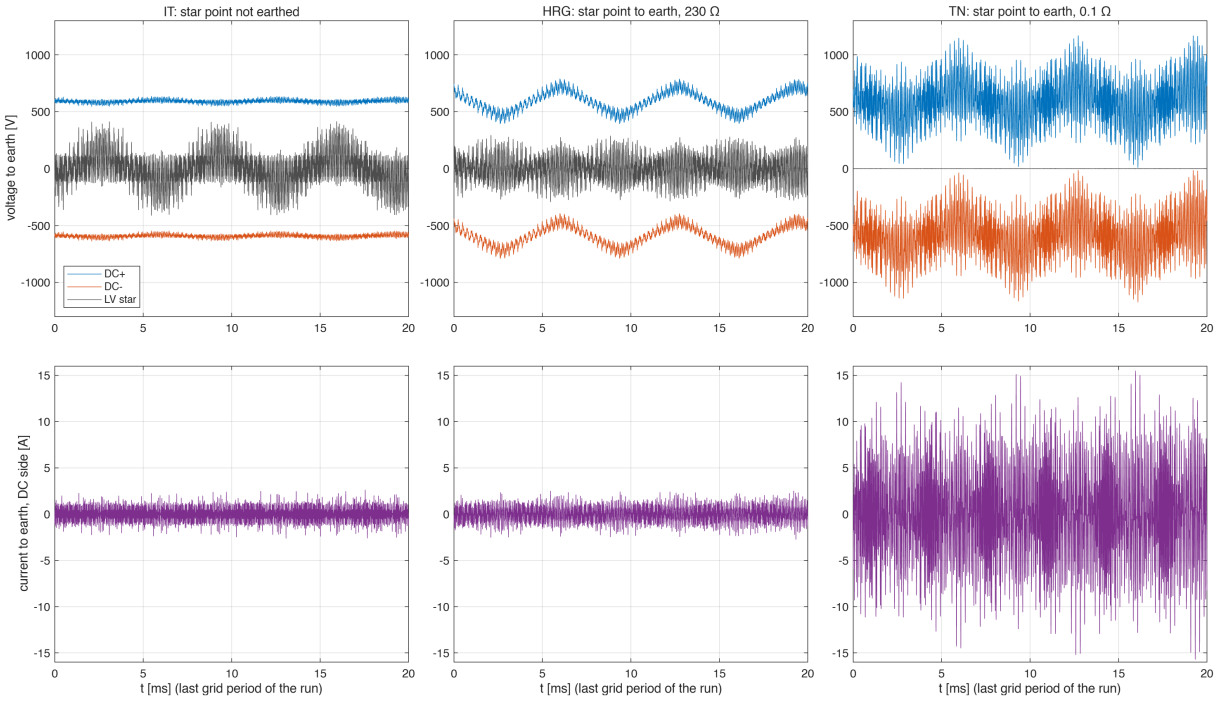


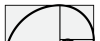
Figure 6: The three earthing schemes, last grid period of the run: voltages to earth of DC+, DC− and of the LV star point (top), current to earth of the DC side (bottom). Script `earthing_cases.m`.

- **IT.** The battery poles move by 91 V peak to peak, mostly at the switching frequency and in the ringing of the common-mode loop; at 150 Hz the battery has 10 V, i.e. the 9.1 % of the capacitive divider, and the LV star point carries the rest. The current to earth, 0.64 A rms, is almost all above 2 kHz.
- **HRG.** Below the corner frequency the star point is held to earth: the 150 Hz common mode moves entirely onto the battery (112 V, poles 410 V peak to peak), while the HF still divides on the capacitances. The resistor carries 0.46 A rms of common-mode current (49 W): it counts in its rating, and the first-fault detection must tell it from a fault current.
- **TN.** The battery carries the whole common-mode voltage of the bridges: the poles swing by 1158 V peak to peak, with steps of $u_{\mathrm{dc}}/3 = 394$ V at every commutation that stress the insulation between the cells and the racks and cooling plates (against 36 V steps in the IT case). The current to earth is 4.4 A rms (15.7 A peak), seven times the IT case, through the battery insulation and the earth conductors.

The same current flows on the DC side and on the AC side, since it goes from one to the other through earth. The values above 20 kHz contain the ringing of the common-mode loop, resolved with about ten solver steps per period: they are indicative.

7.4. Insulation monitoring (IMD)

In an IT system the first insulation fault does not trip anything: it must be signalled and removed before a second fault on another pole or phase turns it into a short circuit. The insulation monitor (IEC 61557-8 [29]) is connected between the system and earth, superimposes a measuring signal and computes the insulation resistance. For a BESS it must be of the type



for *IT AC systems with galvanically connected DC circuits*, since the LV winding, the bridges and the battery are one system, and it must tolerate the leakage capacitance of the system (here 0.33 µF; several microfarads for a large plant) and the common-mode voltage of the converters. One IMD per galvanically connected system: two IMDs on the same system disturb each other.

The symmetric earthing resistors of the battery and the IMD do the same job in two ways: the resistors fix the potential of the poles and let a simple voltage measurement detect the first fault (a fault on DC+ brings DC+ to earth potential and DC− to $-u_{\mathrm{dc}}$); the IMD measures the insulation directly. With both, the IMD sees $r_{\mathrm{gnd}}/2$ as a permanent insulation fault, so the resistors must be much larger than the IMD threshold, or left out. In the model $r_{\mathrm{gnd}} = 100\,\mathrm{k\Omega}$ gives the poles a defined potential; the real value is a choice of the protection concept.

7.5. Residual current monitoring unit (RCMU)

What it is. An RCMU measures the *residual current*: the sum of the currents of all live conductors of a circuit, which is zero unless part of the current returns through earth. The conductors pass together through one magnetic core (a core-balance or summation current transformer); the core sees only their sum,

$$i_{\Delta} = \sum_{k} i_k = -\,i_{\mathrm{earth}}, \tag{7.1}$$

three phases on the AC side, DC+ and DC− on the DC side. Unlike an RCD, which opens its own contacts, the RCMU is a measuring function inside the converter: its signal goes to the control and protection of the converter, which opens the converter contactor. With power converters it must be of *type B*: it measures smooth DC and AC residual currents up to the kilohertz range (a flux-gate sensor), because an insulation fault behind a rectifying bridge gives a DC residual current that saturates the core of an ordinary (type A) device.

Why a BESS needs it. The IMD finds the slow degradation of the insulation; the RCMU finds the faults that appear at once, as a step of the residual current, and it measures what the IMD cannot: the capacitive leakage current that the common-mode voltage drives through the parasitic capacitances. For the transformerless PV inverters the rule is in IEC 62109-2 and VDE V 0126-1-1 [30, 31], and it is often taken over for battery converters: disconnection when the continuous residual current exceeds 300 mA (converters up to 30 kVA) or 10 mA per kVA (larger converters), and on sudden steps of 30, 60 and 150 mA within 0.3, 0.15 and 0.04 s. The step thresholds catch a fault current added on top of the capacitive leakage, whatever its size.

Where. One summation transformer on the AC side of each PCS (the three phases, the star point of the IT system is not a conductor of the circuit), and one on the DC cables of each battery string (DC+ and DC−); Figure 7. The AC one sees every earth current of the PCS; the DC one tells which string leaks.

What the model says. The currents to earth of Table 6 are the capacitive residual current that an RCMU of this plant sees without any fault; the AC and the DC summation transformers see the same current. Below 2 kHz (type-B devices are specified up to about 1 kHz) it is 3 mA in the

IT scheme and 24 mA with the star point earthed, far from the continuous threshold (2.8 A for a module of 278 kVA). Most of it is above the switching frequency: 0.20 A in the 2–20 kHz band in the IT scheme against 3.9 A in the TN scheme, which matters for devices with an extended bandwidth and for the EMC of the plant. In the earthed schemes the low-frequency part grows with the capacitance of the battery (ten times for some MWh), in the IT scheme it stays limited by the small capacitance of the AC side. A fault can be simulated by a resistor with a switch from one pole to earth: the step of i_Δ is the quantity to compare with the thresholds.

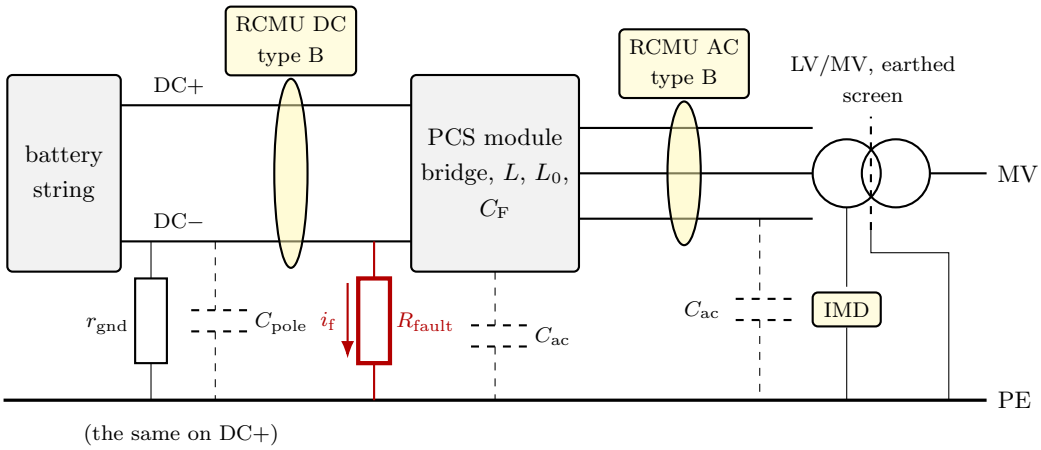


Figure 7: RCMU and IMD in a BESS with an IT LV system (one PCS and one string shown). The summation transformers see the sum of the currents of their conductors, i.e. the current that returns through earth: the capacitive leakage through C_{pole} and C_{ac} (dashed) and the current of an insulation fault (red). The IMD sits between the LV system and earth; the screen of the transformer keeps the common-mode current away from the MV network.

8. Three-level modules (future)

A three-level module (NPC or T-type) changes the common mode and the earthing in the right direction and adds one constraint:

- the common-mode steps are $u_{dc}/6$ instead of $u_{dc}/3$: half the dv/dt seen by the parasitic capacitances, half the HF earth current for the same switching speed, smaller common-mode chokes;
- each module has a neutral point in its split DC link; one high-resistance earthing at a neutral point gives the reference of the IT system without the two pole resistors;
- the neutral-point balance is done with the zero-sequence voltage or with the redundant small vectors: in modules on one battery this changes the zero-sequence voltage of each module and drives the circulating current of Section 6. The balancing of the modules must be coordinated (same strategy, aligned carriers) or the circulating current controlled;
- the control V1 does not change; the library already has a three-level T-type power stage as a Simscape component (`+afe_ttype_ssc/afe_three_level_ttype_lcl.ssc`), so the step is the power stage, the modulator and the neutral-point loop.



9. Conclusions and next steps

The model now has the parasitic capacitances and the earthing resistances of an industrial BESS, with the LV side not earthed and the battery referenced to earth through high resistances; the three earthing schemes can be compared with one parameter, and the simulation confirms that the battery stays quiet only when the LV star point is not held to earth. For the power sharing the control V1 is a droop control equivalent to a virtual synchronous machine, with a virtual impedance in front of a voltage loop on the filter capacitor and of the existing current loop: it shares the active power exactly, works the same on the grid and in the island, and needs the PTP time already modelled to remove the effect of the clock errors on the sharing. It is the droop mode of the library block `grid_forming_dqctrl`; on the companion model `pgfm_droop` it passes the tests T0 to T5 and the black start, with the frequencies, voltages and sharing errors that the analysis predicts. The next steps are:

1. breaker and local load in `pgfm_ctrl_analysis` and the control V1 in its modules (route B), with the common-mode currents of the modules logged;
2. version 2: secondary control of frequency and voltage, a slow correction of the reactive sharing by the plant controller, resynchronization, fault ride-through as grid-forming units (test T6), circulating-current loop;
3. a battery sized for the three modules (and its parasitic capacitance), the initial voltage of the DC links;
4. three-level modules.



A. Files

Table 7: Files referred to in this note.

File	Content
`grid-forming`, **folder** `parallel_of_GFMs/`	
`init_model.m`	initialization, one `param` struct; `param.gnd` (Table 5); puts the folder of this note on the path
`pgfm_ctrl_analysis.slx`	the model; `battery_1/HF and Ground Network`, `grid/grid_emulator/3ph_Dyg_ssc` (`R_star`, `C_lv_earth`); the annotation *Design note* at the top left opens this PDF
`earthing_cases.m`	the three earthing schemes of Section 7.3: table and figure
`power_sharing/init_pgfm_droop.m`	parameters and cases of the companion model (Section 5.13)
`power_sharing/build_pgfm_droop.m`	builds `pgfm_droop.slx`, the companion model of route A: change the script, not the model
`power_sharing/pgfm_droop_tests.m`	runs the cases, checks them and writes `results/<case>.txt`, `.png`
`library`	
`masked_models/grid_forming/`	block `grid_forming_dqctrl` and its control `private/afe_dq_ctrl_pwm.m` with the droop mode (`p.ctrl.droop = 1`); `grid_forming_param_example.m` with the fields `droop_*`; `test/grid_forming_droop_test_run.m`
`foundation/pwr_electronics/+threephase_inductor/`	three-phase inductor: L_0 is the zero-sequence inductance per phase
`foundation/pwr_electronics/+afe_ttype_ssc/`	three-level T-type power stage (Section 8)
`repositories-documentation`, **folder** `repos_doc/parallel_of_gfms/`	
`parallel_of_gfms.tex`, `.pdf`	this note
`figures/hf_cases.png`	Figure 6, written by `earthing_cases.m`
`figures/pgfm_droop_T3.png`, `_T5.png`	Figures 3 and 4, written by `pgfm_droop_tests.m`

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